



TigerAnalytics

AWS Data Engineer Interview Questions

Most-Asked Topics from Real Interviews



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Core Spark Questions

1. What is the difference between **RDD** and **DataFrame** in Spark?
2. How do you optimize for **memory and cost** in ETL Spark jobs?
3. Explain **wide vs. narrow transformations** and their performance impact.



SQL & Analytics Logic

- 4. How to find the **second-highest salary** using window functions?
- 5. Difference between **ROW_NUMBER()**, **RANK()**, **DENSE_RANK()**?
- 6. Write unit tests for **PySpark code** locally?



Spark Architecture & Deployment

- 7. Role of **coalesce()** vs **repartition()** in Spark?
- 8. How do you **deploy PySpark** on EMR or AWS Glue?
- 9. How to **verify production job changes** post-deployment?



Data Products on AWS

10. How to **create data products** on AWS?

11. How does **SparkSession** work in Glue and role of **GlueContext**?

12. How does **data distribution & partitioning** affect Spark performance?



AppFlow & Alternatives

- 13. What is **AWS AppFlow** and where is it used?
- 14. Is AppFlow scalable with 100+ flows?
- 15. Alternatives like **AWS Glue** or **Step Functions**?



API Integration and File Format

- 16. In Glue, how to pull data from **APIs** securely?
- 17. How does **Secrets Manager** help in managing API credentials?
- 18. Why choose **Parquet format** for storage?
- 19. What is **run-length encoding** in Parquet and why is it useful?



Spark Internals & Debugging

- 20. Difference between **Transformations** vs **Actions** in Spark?
- 21. Why does Spark use **lazy evaluation**?
- 22. How to identify and fix a **memory leak** in production?



Real-World Coding Scenarios

- 23. Split full name into first and last name using **PySpark**
- 24. Get highest salary in each department (SQL)
- 25. Find employees with salary **greater than their manager**



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